

4.1 Project Planning

1. Project Implementation Roadmap & WBS Schematic

The diagram below details the 5-phase Work Breakdown Structure (WBS), milestone timelines, resource allocations, and risk management framework for deploying the SCMS application on ServiceNow.

2. Work Breakdown Structure (WBS) & Sprint Schedule

Phase / Milestone	Target Timeline	Core Deliverables & Tasks	Primary Technical Resource
Phase 1: Foundation & Data Model	Weeks 1 – 2	Setup scoped application container (<code>x_scms</code>), schema creation (<code>x_scms_student</code> , <code>x_scms_faculty</code> , <code>x_scms_course</code> , <code>x_scms_leave_request</code> , <code>x_scms_complaint</code>), configure primary/foreign key relationships.	ServiceNow Application Developer
Phase 2: Business Logic & Validation	Weeks 3 – 4	Implement client scripts (<code>onChange</code> date checks), UI policies, server-side Business Rules (duplicate enrollment prevention), and Script Includes (<code>SCMSUtils</code>).	ServiceNow Developer
Phase 3: Automation & Approvals	Weeks 5 – 6	Build multi-tier Flow Designer flows (Faculty Advisor $\rightarrow$ HOD leave approvals), set up auto-assignment rules for campus complaints, configure email notification templates.	Workflow / Automation Specialist

Phase 4: Portal & Catalog Development	Weeks 7 – 8	Construct Service Portal Record Producers for Leave Requests and Facility Complaints, set up user roles (x_scms.student, x_scms.faculty, x_scms.staff), configure granular ACL security rules.	Portal Developer / System Admin
Phase 5: Testing, Analytics & Deployment	Weeks 9 – 10	Conduct UAT, build real-time executive reporting dashboards, export Application Update Sets, finalize capstone technical documentation.	Technical Lead / QA Lead

3. Risk Assessment & Mitigation Matrix

- **Risk 1: Escalation Delays in Approval Workflows**
 - *Impact:* High | *Likelihood:* Medium
 - *Mitigation:* Configure 48-hour auto-escalation timers in Flow Designer to route stagnant requests directly to the Department Head or Dean.
- **Risk 2: Unauthorized Data Access Across Departments**
 - *Impact:* High | *Likelihood:* Low
 - *Mitigation:* Enforce strict row-level Access Control Lists (ACLs) restricting students to self-owned records and faculty to department-scoped tables.
- **Risk 3: Data Loss During Instance Upgrades/Reclamation**
 - *Impact:* Critical | *Likelihood:* Medium
 - *Mitigation:* Commit all application changes daily into Source Control (Git) and export Update Sets at the end of each sprint.